



Problem Sets for Macroeconomics

March, 2026

PS 1

1.1 (计算题). The nation of Narnia has macroeconomic data, as shown in the following table. Suppose that there is no statistical discrepancy and that net factor payments include only compensation from the rest of the world.

- Calculate Narnian GDP using the expenditure approach and the income approach.
- Find gross national product, net national product, national income.
- Find gross national factor income and net national factor income.
- Find personal income and disposable personal income.

Table 1.1: Data for PS 1.1 (Billions of dollars)

1	Personal consumption expenditures	3000
2	Gross private domestic investment	2500
3	Government consumption expenditures and gross investment	1000
4	Net exports of goods and services	500
5	Indirect business taxes less subsidies	600
6	Business transfers, domestic	150
7	Current surplus of government enterprises (always domestic)	50
8	Proprietors' income, domestic	300
9	Rental income of persons, domestic	400
10	Net Interest, domestic	900
11	Corporate profits, domestic	1400
12	Compensation of employees, domestic	3000
13	Consumption of fixed capital (always domestic)	200
14	Net factor payments (wages from the rest of the world)	100
15	Personal interest income	800
16	Personal dividend income	1200
17	Personal current transfer receipts from government and business	250
18	Contributions for government social insurance, domestic	350
19	Personal income taxes	1000

1.2. The nation of Narnia is comprised of a coal producer, a steel producer, and some consumers (there is no government). In a given year, the coal producer produces 15 million tons of coal and sells it for \$5 per ton. The coal producer pays \$50 million in wages to consumers. The steel producer uses

25 million tons of coal as an input into steel production, all purchased at \$5 per ton. Of this, 15 million tons of coal come from the domestic coal producer and 10 million tons are imported. The steel producer produces 10 million tons of steel and sells it for \$20 per ton. Domestic consumers buy 8 million tons of steel, and 2 million tons are exported. The steel producer pays consumers \$40 million in wages. All profits made by domestic producers are distributed to domestic consumers. There is no depreciation.

- a. Determine Narnian GDP using (i) the product approach, (ii) the expenditure approach, and (iii) the income approach.
- b. Determine Narnian current account surplus.
- c. Find GNP of Narnia.
- d. Determine GNP and GDP in the case where the coal producer is owned by foreigners, so that the profits of the domestic coal producer go to foreigners and are not distributed to domestic consumers.

1.3. Suppose that the government does not make investment. Consider an identity $S^{\text{Prv}} - I = CA + D$, where S^{Prv} is private sector saving, I is gross private domestic investment, CA is the current account surplus, and D is the government deficit.

- a. Show that the above identity holds.
- b. Explain what the identity means.

1.4. Chain-weighted growth rate of real GDP and quality improvement. In year 1 and year 2, there are two products produced in a given economy, computers and bread. Suppose that there are no intermediate goods. In year 1, 20 computers are produced and sold at \$1,000 each, and in year 2, 25 computers are sold at \$1,500 each. In year 1, 10,000 loaves of bread are sold for \$1.00 each, and in year 2, 12,000 loaves of bread are sold for \$1.10 each.

- a. Calculate the percentage increase in real GDP from year 1 to year 2 using the chain-weighting method.
- b. Calculate the percentage inflation rate from year 1 to year 2 using the chain-weighting method.
- c. Suppose that computers in year 2 are twice as productive as computers in year 1. That is, computers are of higher quality in year 2 in the sense that one computer in year 2 is equivalent to two computers in year 1. How does this change your calculations in parts (a) and (b)?

1.5 (分析题). House prices and bubbles. Houses can be thought of as assets with a fundamental value equal to the expected present discounted value of their future real rents.

- a. Would you prefer to use real payments and real interest rates or nominal payments and nominal interest rates to value a house?

Suppose that the real rent R on a house received at the end of each period, the real interest rate r , and the real risk premium $\tilde{\alpha}$ are constant for ever. Let Q_t be the real price of a house at the beginning of period t . The rent on a house is like the dividend on a stock. Suppose Q_t is equal to the real fundamental value of a house.

- b. Find the real price of a house.
 c. If the interest rate falls, what will happen to the price-to-rent ratio?
 d. If houses are perceived as a safer investment, what will happen to the price-to-rent ratio?

PS 2

2.1. Consider a Cobb-Douglas production function with three inputs. K is capital (the number of machines), N is labor (the number of workers), and H is human capital (the number of college degrees among the workers). The production function is $Y = K^{1/3}N^{1/3}H^{1/3}$.

- a. Derive an expression for the marginal product of labor. How does an increase in the amount of human capital affect the marginal product of labor?
 b. Derive an expression for the marginal product of human capital. How does an increase in the amount of human capital affect the marginal product of human capital?
 c. What is the income share paid to labor? What is the income share paid to human capital? In the national income accounts of this economy, what share of total income would workers appear to receive?
 d. An unskilled worker earns the marginal product of labor, whereas a skilled worker earns the marginal product of labor plus the marginal product of human capital. Using your answers to parts (a) and (b), find the ratio of the skilled wage to the unskilled wage. How does an increase in the amount of human capital affect this ratio?
 e. Some people advocate government funding of college scholarships as a way of creating a more egalitarian society. Others argue that scholarships help only those who are able to

go to college. Do your answers to the preceding questions shed light on this debate?

2.2. A *tax credit* is a sum subtracted from the total amount that a taxpayer owes to the state (see Wiki's [tax credit](#)). When the government subsidizes investment by an investment tax credit, the subsidy often applies to only some types of investment. This question asks you to consider the effect of such a change. Suppose there are two types of investment in the economy: business investment and residential investment. Public investment is zero. Suppose that the government institutes an investment tax credit only for business investment. According to the classical macroeconomic theory for a closed economy, answer the following questions.

- a. How does this policy affect the demand curve for business investment? How about the demand curve for residential investment?
- b. Draw the economy's supply of and demand for loanable funds. How does this policy affect the supply and demand for loanable funds? What happens to the equilibrium interest rate?
- c. Compare the old and the new equilibria. How does this policy affect the total quantity of investment? How about the quantity of business investment? How about the quantity of residential investment?

2.3 (实战题). Some economic historians have noted that during the period of the [gold standard](#), gold discoveries were most likely to occur after a long deflation. The discoveries of 1896 ([Klondike Gold Rush](#)) are such examples. Why might this be true?

2.4 (画图题). Let net factor payments (NFP) be fixed at 0. Suppose that there is no government investment. Suppose that the exchange rate floats freely. Suppose that China (Home) is a small open economy and the rest of the world (Foreign) can be regarded as a large open economy. With the aid of graphics, try to predict what would happen to China in response to each of the following events.

- a. A fall in consumer confidence about Chinese future induces Chinese consumers to spend less and save more.
- b. After financial crises, Foreign turns to protectionist trade policies to improve trade deficit.
- c. Foreign begins to subsidize investment by instituting an investment tax credit.

China has become a large open economy recent years. What would happen to China in response to each of the above events.

2.5 (實戰題; 15). 閱讀《管子・卷第十二・侈靡第三十五》節選內容。根據本課程內容，回答問題。

(管仲曰)“地重人載，毀敝而養不足。事末作而民興之，是以下名而上實也。聖人者，省諸本而游諸樂，大昏也，博夜也。”(齊桓公)問曰：“興時化若何？”(管仲曰)“莫善於侈靡。”

如你在考場上睡著了，一覺醒來發現穿越 2600 多年到東周齊桓公身邊，身居相位。你打算向齊桓公提供什麼建議以解決齊國面臨的“毀敝而養不足”問題？

PS 3

3.1. The virtue of thrift in the Solow–Swan Model. Suppose there is no technological progress. If each person saves more in the Solow–Swan model, does the saving per worker rise? What is the effect of an increase in the saving rate on the output per worker?

3.2. Suppose that the economy’s production function is $Y = \sqrt{K}\sqrt{AN}$, that the saving rate, s , is equal to 16%, and that the rate of depreciation, δ , is equal to 10%. Suppose further that the number of workers grows at 2% per year and that the rate of technological progress is 4% per year.

- a. Find the steady-state values of the variables listed below.
 - i. The capital stock per effective worker;
 - ii. Output per effective worker;
 - iii. The growth rate of output per effective worker;
 - iv. The growth rate of output per worker;
 - v. The growth rate of output.
- b. Suppose that the rate of technological progress doubles to 8% per year. Recompute the answers to part (a).
- c. Now suppose that the rate of technological progress is still equal to 4% per year, but the number of workers now grows at 6% per year. Recompute the answers to part (a). Are people better off in part (a) or in part (c)?

3.3. Describe how each of the following events affects the break-even and actual investment curves in the Solow–Swan model:

- a. The rate of depreciation falls.
- b. The rate of technological progress rises.

- c. The production function is $Y = K^\alpha(AN)^{1-\alpha}$ and capital's share, α , rises.
- d. Workers exert more effort, so that output per unit of effective labor for a given value of capital per unit of effective labor is higher than before.

3.4. In the nation of Narnia, the capital share of GDP is about 30 percent, the average growth in output is about 3 percent per year, the depreciation rate is about 4 percent per year, and the capital-output ratio (K/Y) is about 2.5. Suppose that the production function is $Y = K^\alpha(AN)^{1-\alpha}$ and that the nation of Narnia has been in a steady state.

- a. What must the saving rate be in the steady state? What is the marginal product of capital in the steady state?
- b. Suppose that public policy raises the saving rate so that the economy reaches the Golden Rule level of capital. What must the saving rate be to reach the Golden Rule steady state? What will the marginal product of capital be at the Golden Rule steady state? What will the capital-output ratio be at the Golden Rule steady state?

3.5. The amount of education that a typical person receives varies substantially among countries. Suppose you were to compare Westland with a highly educated labor force and Eastland with a less educated labor force. Assume that the production function is given by $Y = F(K, EN)$, where education, denoted by E , affects only the level of the efficiency of labor N . Also assume that the two countries are otherwise the same: they have the same saving rate s , the same depreciation rate δ , the same population growth rate n , and the same growth rate g of education. Both countries are described by the Solow–Swan model and are in their steady states. Compare the following variables between two countries?

- a. The rate of growth of total income.
- b. The level of income per worker.
- c. The real rental price of capital.
- d. The real wage per worker.

3.6. For a neoclassical production function $Y = F(K, AN)$, show that each competitive factor of production earns its marginal product in the Solow–Swan model. Show that if all capital income is saved and all labor income is consumed, then the economy reaches the golden rule level of capital accumulation.

3.7. **Piketty** (2014, p.572) finds that the average rate of return on capital is about 4 or 5 percent per year while the

growth rate of output will not exceed 2 percent per year in the long run. Piketty (2014, p.571) concludes that if the private rate of return on capital is greater than the rate of growth of output, capital will grow faster than output, and K/Y will increase without bound. Workers who own nothing but labor will become more and more dominated by those who own capital stock. Is it possible for Piketty to be FALSE in the Solow model?

3.8. In the Solow model without technological progress, let N denote total working age population and let $\xi \in (0, 1)$ denote the participation rate. ξN is the labor force in the Solow economy. Suppose the labor force is totally employed, which is consistent with the assumption of vertical labor supply curve. The growth rate of N is $n > 0$. The saving rate is s . The depreciation rate is δ . Suppose the production function is neoclassical and can be written as $Y = F(K, \xi N)$. In the vicinity of the balanced growth path, how rapidly does Y/N converge to the balanced growth path? What are the effects of an increase in ξ on K , Y , and C along the balanced growth path? (Calculate partial derivatives)

3.9. Suppose the production function is given by $Y = K^\alpha N^{1-\alpha}$, $\alpha \in (0, 1)$. The fundamental equation of the Solow model is changed into

$$\dot{K}(t) = A(t) \cdot sY(t) - \delta \cdot K(t).$$

The depreciation rate of K is $\delta \in (0, 1)$. The growth rates of technology, A , and the number of workers, N , are g and n , respectively. What are the growth rates of Y and K in steady state?

3.10. Alter the Solow–Swan growth model so that the production technology is given by $Y = AK$, where Y is output, K is capital, and A is technology which is constant. Thus, output is produced only with capital.

- Show that it is possible for income per person to grow indefinitely.
- Also show that an increase in the saving rate increases the growth rate of per capita income.
- From parts (a) and (b), what are the differences between this model and the Solow–Swan model without technological progress?

3.11. Government expenditure financed by a lump-sum tax. In the Solow–Swan model without technological progress, the government purchases G units of consumption goods in each period. The government finances its purchases G through lump-sum taxes on workers. Let N denote the number of workers. Each worker pays a lump-sum tax $\tau > 0$. Suppose that the government budget is balanced each period, that is, $G = T = \tau N$. Consumers consume a constant fraction of disposable income—that is, $C = (1 - s)(Y - T)$, where $s \in (0, 1)$ is the saving rate.

- Determine the steady state of capital per worker, k^* .
- Show that there can be two steady states, one with high k^* and the other with low k^* .
- Ignore the steady state with low k^* (it can be shown that this steady state is “unstable”). Determine the marginal effects of an increase in τ on capital per worker and on output per worker in the steady state. What are the marginal effects of $\tau \uparrow$ on the steady-state growth rates of aggregate output, aggregate private consumption, and aggregate investment?

3.12. Government expenditure financed by a proportional tax. In the Solow–Swan model without technological progress, the production function is neoclassical: $Y(t) = F[K(t), N(t)]$, where Y, K , and N are the level of output, physical capital, and labor input, respectively. Suppose that the government always keeps a balanced budget constraint, that is, government expenditure $G(t)$ is equal to the tax revenue $T(t)$ for all t . Suppose that the tax revenue is a constant share of income $Y(t)$ for all t , that is, $T(t) = \tau \times Y(t)$ with a constant proportional tax rate $\tau \in (0, 1)$. The equilibrium in the market for goods is $Y(t) = C(t) + I(t) + G(t)$, where $C(t)$ is the aggregate private consumption and $I(t)$ is the aggregate private investment. Suppose that the consumption is $C(t) = c_1 \times [Y(t) - T(t)]$ where $c_1 \in (0, 1)$ is the constant marginal propensity to consume. Suppose that the government investment is a constant share of $G(t)$ for all t , that is, $G^I(t) = \zeta \times G(t)$ where $\zeta \in [0, 1]$. Let $n > 0$ be the constant growth rate of labor $N(t)$ and $\delta \in (0, 1)$ be the constant depreciation rate of physical capital stock $K(t)$. Find the marginal effect of an increase in τ on the physical capital per worker and the output per worker in the steady state.

3.13. Suppose there are two countries, A and B , and each is a Solow–Swan economy without technological progress. In

both countries, the growth rate of labor is $n > 0$ and the depreciation rate of physical capital is $\delta > 0$. In each country, a fraction a of the population is rich, and a fraction $1 - a$ is poor. Suppose that rich people save a fraction s_r of their income, and poor people save a fraction $s_p < s_r$ of their income, no matter what country they live in. In country A , suppose that rich people as a group receive a fraction χ_A of total income, while in country B rich people as a group receive χ_B fraction of total income. Assume $\chi_A > \chi_B$. If you were a poor person, where would you rather live, in country A or country B ? What if you are rich?

3.14. In an economy, the production function is given by $Y = F(XK, AN)$, where $X(t) = X_0 e^{xt}$, $A(t) = A_0 e^{gt}$, and $N(t) = N_0 e^{nt}$. The law of motion of K is given by $\dot{K} = sY - \delta K$, where $s \in (0, 1)$ is the constant saving rate and $\delta \in (0, 1)$ is the depreciation rate of physical capital stock. Suppose that $F(\cdot, \cdot)$ satisfies $F_1 > 0, F_2 > 0$, and exhibits the property of constant returns to scale. Suppose that the economy has a steady state. Can F always be written as a production function with only labor-augmenting technological progress?

3.15. The growth rate in period t of a discrete-time variable X_t , denoted by χ_t , is defined as $\chi_t = \frac{X_t - X_{t-1}}{X_{t-1}}$. In a discrete-time Solow model of a closed economy, the production function is $Y_t = F(K_t, A_t N_t)$, where Y_t is the level of output in period t , K_t the physical capital stock, N_t the number of workers, A_t the index of technology, and $F(\cdot, \cdot)$ is neoclassical. Suppose that the growth rate of A_t is constant $g > 0$ and that the growth rate of N_t is constant $n > 0$. The dynamic equation for physical capital is $K_{t+1} = K_t + sY_t - \delta K_t$, where $s \in (0, 1)$ is the constant saving rate and $\delta \in (0, 1)$ is the constant rate of depreciation of physical capital. Answer the following questions.

- Show that there exists a balanced growth path along which the output is strictly positive.
- Find the growth rate of output per capita along the balanced growth path.

PS 4

4.1. The paradox of thrift in the Keynesian cross model. Suppose the consumption function is $C = C_0 + c_1 \cdot (Y - \bar{T})$.

If households save more (captured by $C_0 \downarrow$) in the Keynesian cross model, what would happen to saving S ? Should the virtues of thrift be recommended?

4.2. The paradox of thrift in the IS-LM model. In the IS-LM framework, we assume that investment depends negatively on the interest rate and positively on output. That is, $I = b_0 + b_1 Y - b_2 i$. The consumption function is $C = C_0 + c_1 \cdot (Y - \bar{T})$. Suppose the demand for real money balance is $M^d/P = d_1 Y - d_2 i$. Suppose $b_0 > 0, b_1 > 0, b_2 > 0, C_0 > 0, c_1 > 0, d_1 > 0, d_2 > 0$, and $1 - c_1 - b_1 > 0$.

- Show the effect of the fall in consumer confidence (captured by $C_0 \downarrow$) on output and the interest rate.
- How will the fall in consumer confidence affect consumption, investment, and private saving? Will the attempt to save more necessarily lead to more saving?

4.3. Disturbances in the Mundell-Fleming (MF) model. Suppose that China, a small open economy, chooses fixed exchange rate regime and that the rest of the world, a large open economy, always chooses floating exchange rates regime. Net exports do not depend on income. Net factor payments are always 0. Use the MF model to predict effects of each of the following events on Chinese income, nominal exchange rate of CNY and Chinese net exports.

- A fall in consumer confidence about Chinese future induces Chinese consumers to spend less and save more.
- The introduction of a stylish line of Mercedes-Benz makes Chinese consumers prefer foreign cars over domestic cars.
- Introducing automated teller machines (ATM).

Suppose that China turns to floating exchange rate regime. Use the MF model to predict again effects of above events.

China has become a large open economy recent years. What would happen to China in response to each of the above events, under fixed and flexible exchange rates, respectively?

4.4. In years 2014 and 2015, Jilin province experienced a recession relative to the nation. Suppose that provinces are allowed to print CNY, and that Jilin is a small economy relative to the nation.

- If Jilin province would try to improve its employment, would you recommend monetary expansion or fiscal expansion? Give your explanation.
- If Jilin province turned to local protectionism which prohibited local firms from buying goods produced by other

provinces, what would happen to the output and trade balance of Jilin province?

4.5. The Mundell-Fleming (MF) model takes the world interest rate i^* as an exogenous variable. Let's consider what happens when this variable changes.

- What might cause the world interest rate to rise? (Hint: The world is a closed economy.)
- In the Mundell-Fleming model with floating exchange rates, what happens to aggregate income Y , the exchange rate e , and the trade balance $NX(eP/P^*)$ when the world interest rate rises?
- In the Mundell-Fleming model with a fixed exchange rate, what happens to aggregate income Y , the nominal exchange rate e , and the trade balance $NX(eP/P^*)$ when the world interest rate rises?

4.6. Suppose that in the Mundell-Fleming (MF) model, the demand for real money balances depends on disposable income $Y - T$, so that the equation for the money market becomes $M/P = L(i, Y - T)$. Analyze the short-run impact of a tax cut in a small open economy on the exchange rate and income under floating and fixed exchange rates, respectively.

4.7. Suppose Home is a small open economy. Suppose that the price level relevant for money demand includes the price of imported goods and that the price of imported goods depends on the exchange rate. That is, the money market is described by $M/P = L(i, Y)$ and $P = \lambda P_D + (1 - \lambda)P_F/e$ where P_D denotes the price of domestic goods, P_F the price of foreign goods measured in the foreign currency, e the nominal exchange rate, λ the share of domestic goods in the price index P . Assume that P_D and P_F are sticky in the short run.

- Draw Home's LM curve for given values of P_D and P_F in the space (Y, e) . Is it still vertical?
- What is the effect of Home's expansionary fiscal policy under floating exchange rates?
- What is the effect of Home's expansionary monetary policy under floating exchange rates?

4.8. In a small open economy with imperfect capital mobility and floating nominal exchange rates, the IS relation is given by $Y = C(Y - T) + I(r) + G + NX(\epsilon)$, where T is the lump-sum tax, r is the real interest rate, ϵ is the real exchange rate, $C'(\cdot) \in (0, 1)$, $I' < 0$, $NX'(\cdot) < 0$. The LM

relation is given by $M/P = L(Y, r + \pi^e)$, where π^e is the expected rate of domestic inflation, $L_Y > 0$, $L_r < 0$. Suppose that there is no government intervention. Imperfect capital mobility can be described by $NX(\epsilon) = NFI(r - r^*)$, where $\frac{d}{d(r-r^*)}NFI(r - r^*) \in (-\infty, 0)$. What are the effects of an increase in π^e on the equilibrium levels of Y, r , and ϵ ? (Calculate partial derivatives)

PS 5

5.1. The sticky-wage model. Suppose the production function is $Y = 10\sqrt{N}$, where N is the labor input.

- Find the demand for labor.
- In the classical model, suppose that, in equilibrium, the nominal wage rate is $W = 2$ and the level of price is $P = 1$. Find the potential level of output.
- In the sticky-wage model, suppose the nominal wage rate is fixed at $W = 2$ by contract in the short run. Find the AS relation.

5.2. The paradox of thrift in the AS-AD model. Suppose the economy begins with output equal to its natural level. Then there is a decrease in consumer confidence (captured by $C_0 \downarrow$) as households attempt to increase their saving for a given level of disposable income.

- What happens to output, the interest rate, the price level, consumption, investment, and private saving in the short run? Is it possible that the decline in consumer confidence will actually lead to a fall in private saving in the short run?
- Is there any paradox of saving in the long run?

5.3. Suppose the money demand curve is flat when the interest rate i is equal to zero and downward-sloping as usual when the interest rate i is greater than zero.

- Draw the IS, LM and AD curves.

Suppose that the economy starts from a short-run equilibrium where output is below the natural level of output. Based on your answer to part (a), answer the following questions.

- If the central bank increases the money stock, what will be the effects on output and price level in the short run and the long run?
- If the government raises government purchases, what will be the effects on output and price level in the short run and the long run?

5.4. Pigou Effect. Pigou (1943) said, falling prices would raise real wealth, which would also raise consumption expenditure. Thus the economy will not get stuck in the liquidity trap and automatically return to full employment. Suppose that the consumption is $C = C(Y - T, A)$, where $Y - T$ is the disposable income and A is the real wealth. Suppose $0 < \partial C / \partial Y < 1$ and $\partial C / \partial A > 0$. The nominal wealth includes bonds (B) and money (M). The real wealth is given by $A = (B + M) / P$. Is it possible for Pigou to be right?

5.5. In a closed AS–AD model, the AD relation is given by $Y = 700 + 0.5 \frac{M}{P}$ while the AS relation is given by $Y = 600(P - P^e) + 1000$, where Y is the level of output, P the actual price level, P^e the expected price level, and M the money supply. Let P and Y be positive. Suppose that an economy starts from $M = 600$ and $P^e = P$. Consider the effect of monetary expansion from $M^S = 600$ to $M^S = 3600$.

- If the central bank makes an announcement of the monetary expansion, find Y and P in the short-run equilibrium.
- If the central bank increases the money supply *secretly* such that people do not know the monetary expansion, find Y and P in the short-run equilibrium.

5.6. A central bank has decided to adopt inflation targeting and is now debating whether to target 5 percent inflation or zero inflation. The economy is described by the following Phillips curve: $u = 5 - 0.5(\pi - \pi^e)$, where u and π are the unemployment rate and inflation rate measured in percentage points (i.e., $u = 5$ means the unemployment rate is 5%). The social cost of unemployment and inflation is described by the following loss function: $L = u + 0.05\pi^2$. The central bank would like this loss to be as small as possible.

- If the central bank commits to target 5 percent inflation, what is expected inflation? If the central bank follows through, what is the unemployment rate? What is the loss from inflation and unemployment?
- If the central bank commits to target zero inflation, what is expected inflation? If the central bank follows through, what is the unemployment rate? What is the loss from inflation and unemployment?
- Based on your answers to parts (a) and (b), which inflation target would you recommend? Why?
- Suppose the central bank chooses to target zero inflation, and expected inflation is zero. Suddenly, however, the central bank surprises people with 5 percent inflation. What

- is unemployment in this period of unexpected inflation?
 What is the loss from inflation and unemployment?
 e. What problem does your answer to part (d) illustrate?

5.7. Inflation targeting and the interest rate rule in the IS-LM model. Consider a closed economy in which the central bank follows an interest rate rule. The IS relation is given by $Y = C(Y - T) + I(Y, r) + G$, where r is the real interest rate, $C' \in (0, 1)$, $I_Y > 0$, $I_r < 0$, and $I_Y < 1 - C'$. The central bank sets the nominal interest rate i according to the rule

$$i = i^* + a(\pi^e - \pi^*) + b(Y - Y_n) \quad (5.1)$$

where π^e is expected inflation, π^* is the target rate of inflation, and Y_n is the natural level of output. Assume that $a > 1$ and $b > 0$. The symbol i^* is the target interest rate the central bank chooses when expected inflation equals the target rate and output equals the natural level. The central bank will increase the nominal interest rate when expected inflation rises above the target, or when output rises above the natural level. The rule (5.1) is slightly different from the Taylor rule in that the rule (5.1) uses expected inflation instead of actual inflation. Real and nominal interest rates are related by the Fisher equation $r = i - \pi^e$.

- a. Let $r^* = i^* - \pi^*$. Use the definition of the real interest rate to express the interest rate rule (5.1) as

$$r = r^* + (a - 1)(\pi^e - \pi^*) + b(Y - Y_n). \quad (5.2)$$

- b. Graph the IS relation in a diagram, with r on the vertical axis and Y on the horizontal axis. In the same diagram, graph the interest rate rule (5.2) for given values of π^e, i^*, π^*, Y_n . Call the interest rate rule the monetary policy (MP) relation.
- c. Using the diagram you drew in part (b), show that an increase in government spending leads to an increase in output and the real interest rate in the short run.
- d. Now consider a change in the monetary policy rule. Suppose the central bank reduces its target inflation rate, π^* . How does the fall in π^* affect the MP relation? What happens to output and the real interest rate in the short run?
- e. Suppose the economy starts with $Y = Y_n$ and $\pi^e = \pi^*$. Now suppose there is an increase in π^e . Assume that Y_n does not change. Using the diagram you drew in part (b), show how the increase in π^e affects the MP relation. What happens to output and the real interest rate in the short run?

- f. Without attempting to model the dynamics of inflation explicitly, assume that inflation and expected inflation will increase over time if $Y > Y_n$, and that they will decrease over time if $Y < Y_n$. Given the effect on output you found in part (e), will π^e tend to return to the target rate of inflation, π^* , over time?
- g. Redo part (e), but assuming this time that $a < 1$. How does the increase in π^e affect the MP relation when $a < 1$? What happens to output and the real interest rate in the short run?
- h. Again assume that inflation and expected inflation will increase over time if $Y > Y_n$, and that they will decrease over time if $Y < Y_n$. Given the effect on output you found in part (g), will π^e tend to return to the target rate of inflation, π^* , over time? Is it sensible for the parameter a (in the interest rate rule) to have values less than 1?

5.8. In a closed AS–AD model, the IS relation is given by $Y = C(Y - T) + I(i - \pi^e) + G$, where T is the lump-sum tax, i is the nominal interest rate, and π^e is the expected rate of inflation. The LM relation is given by $M/P = L(Y, i)$. The production function is given by $Y = F(K, N)$, where K is the constant capital and N is labor input. Suppose F satisfies $F'_N > 0$ and $F''_{NN} < 0$. Profit maximization gives the demand for labor: $W/P = F'_N(K, N)$, where W is the nominal wage rate. If W is rigid in the short run, find the short-run effect of an increase in G on the equilibrium level of Y . (Calculate partial derivatives)

($\bar{X}$: undisturbed X . Y_n : the potential output)

1 (PS1.1). a. GDP=7000; b. GNP=7100, NNP=NI=6900; c. GNFI=6300, NNFI=6100; d. PI=5700, DPI=4700.

2 (PS1.2). a. GDP=\$150 million; b. CA=-\$10 m; c. GNP=\$150 m; d. GNP = GDP + NFP = 150-25 = \$125 m.

3 (PS1.3). a. Easy; b. Private saving, which is not used to finance domestic investment, is either lent to the domestic government to finance its deficit, or lent to foreigners.

4 (PS1.4). a. 23.495828%; b. 36.846728%; c. 100.748599%, -15.815104%.

5 (PS1.5). a. Real; b. $Q_t = \frac{R\beta}{1-\beta}$, $\beta = \frac{1}{1+r+\alpha}$; c. $\uparrow$; d. $\uparrow$.

6 (PS2.1). a. MPN $\uparrow$; b. MPH $\downarrow$; c. 1/3, 1/3, 2/3; d. The ratio = $1+N/H$; e. More egalitarian.

7 (PS2.2). a. Move rightward; unchanged. b. Supply is unchanged; demand shifts rightward; $r \uparrow$. c. Unchanged; rise; fall.

8 (PS2.3). Easy.

9 (PS2.4). Small Home: a. $\bar{r}$, ϵ depreciating, $NX \uparrow$, $\bar{Y}$, $C \downarrow$. b. $\bar{r}$, ϵ depreciating, $\bar{N}X$, $\bar{Y}$, $\bar{C}$. c. $r \uparrow$, $I \downarrow$, ϵ depreciating, $NX \uparrow$. Large Home: a. $r \downarrow$, ϵ depreciating, $NX \uparrow$, $\bar{Y}$, $C \downarrow$. b. $\bar{r}$, $\bar{NFI}$, $\bar{N}X$, ϵ depreciating, $\bar{Y}$, $\bar{C}$. c. $r \uparrow$, $I \downarrow$, ϵ depreciating, $NX \uparrow$.

10 (PS 2.5). T 同学出的期末考试题。

11 (PS3.1). The saving per worker rises. At the time when the saving rate rises, the output per worker does not change. But the output per worker is higher in the new steady state.

12 (PS3.2). a. (i) $\hat{k}^* = 1$; (ii) $\hat{y}^* = 1$; (iii) 0; (iv) 4%; (v) 6%. b. (i) $\hat{k}^* = 0.64$; (ii) $\hat{y}^* = 0.8$; (iii) 0; (iv) 8%; (v) 10%. c. (i) $\hat{k}^* = 0.64$; (ii) $\hat{y}^* = 0.8$; (iii) 0; (iv) 4%; (v) 10%. People in (a) are better off.

13 (PS3.3). a. The break-even line is flatter. b. The break-even line is steeper. c. The actual investment line is lower in $(0, 1)$ and higher in $(1, +\infty)$. d. The actual investment line is higher.

14 (PS3.4). a. $s = 0.175$; $MPK = 0.12$; b. $s_{gold} = 0.3$, $MPK = 0.07$; $K/Y = \frac{0.3}{0.07} \approx 4.2857$.

15 (PS3.5). a. The same. b. Higher in Westland. c. The same. d. Higher in Westland.

16 (PS3.6). $s\hat{y} - (n + g + \delta)\hat{k} = \hat{k}[f'(\hat{k}) - (n + g + \delta)]$.

17 (PS 3.7). It is possible.

18 (PS 3.8). $\lambda = (1 - \alpha_K)(\delta + n)$. $\frac{\partial K^*}{\partial \xi} = \frac{K^*}{\xi}$. $\frac{\partial Y^*}{\partial \xi} = \frac{Y^*}{\xi}$. $\frac{\partial C^*}{\partial \xi} = \frac{(1-s)Y^*}{\xi}$.

19 (PS 3.9). $\dot{K}/K = \frac{1}{1-\alpha}g + n$. $\dot{Y}/Y = \frac{\alpha}{1-\alpha}g + n$.

20 (PS3.10). $\dot{y}/y = sA - (n + \delta)$. If $sA > n + \delta$, y grows indefinitely.

21 (PS 3.11). a. The steady state is determined by $sf(k) = s\tau + (n + \delta)k$. c. $\tau \uparrow \Rightarrow k_{High}^* \downarrow, y^* \downarrow$. The steady-state growth rates of Y^* , C^* , and I^* are n , independent of τ .

22 (PS3.12 ♡). If $\zeta \in [0, 1 - c_1]$, $\tau \uparrow \Rightarrow k^* \downarrow, y^* \downarrow$. If $\zeta \in (1 - c_1, 1]$, $\tau \uparrow \Rightarrow k^* \uparrow, y^* \uparrow$.

23 (PS 3.13). The rich would like to live in country A. The poor would like to live in A if $s_r / [(s_r - s_p)\chi + s_p] > 1/\alpha_K$.

24 (PS 3.14). Yes!

25 (PS 3.15). The growth rate of output per capita is g along the balanced growth path.

26 (PS4.1). The saving is undisturbed.

27 (PS4.2). a. $i \downarrow, Y \downarrow$. b. $C_0 \downarrow$ leads to $C \downarrow$; $I \downarrow$ if $b_1 d_2 - b_2 d_1 > 0$, $I \uparrow$ if $b_1 d_2 - b_2 d_1 < 0$; the private saving changes the same as I . Not necessarily.

28 (PS4.3 ♡). Small China (flexible e): a. $\bar{i}, e \downarrow, \bar{Y}, NX \uparrow$; b. $\bar{i}, e \downarrow, \bar{Y}, \bar{NX}$; c. $\bar{i}, e \downarrow, Y \uparrow, NX \uparrow$. Small China (fixed e): a. $\bar{i}, \bar{e}, Y \downarrow, \bar{NX}(\bar{e}), M \downarrow$; b. $\bar{i}, \bar{e}, Y \downarrow, NX \downarrow, M \downarrow$; c. $\bar{i}, \bar{e}, \bar{Y}, \bar{NX}, M \downarrow$.

Large China (flexible e): a. $i \downarrow, e \downarrow, Y \downarrow, NX \uparrow, Y^* \downarrow$; b. $\bar{i}, e \downarrow, \bar{Y}, \bar{NX}, \bar{Y}^*$; c. $i \downarrow, e \downarrow, Y \uparrow, NX \uparrow, Y^* \downarrow$. Large China (fixed e): a. $\bar{i}, \bar{e}, Y \downarrow, \bar{NX}, M \downarrow, \bar{Y}^*$; b. $i \uparrow, \bar{e}, Y \downarrow, NX \downarrow, M \downarrow, Y^* \uparrow$; c. $\bar{i}, \bar{e}, \bar{Y}, \bar{NX}, M \downarrow, \bar{Y}^*$.

29 (PS4.4 ♡). a. Fiscal expansion. b. $Y \uparrow, NX \uparrow$.

30 (PS4.5). a. In the short run, i^* is determined by Foreign's IS and LM curves. In the long run, anything that raises the world's demand for investment or lowers the world's saving will raise i^* . b. $Y \uparrow, e \downarrow, NX \uparrow, I \downarrow, C \uparrow$. c. $Y \downarrow, \bar{e}, \bar{NX}, M \downarrow, I \downarrow, C \downarrow$.

31 (PS4.6). Floating e : $e \uparrow, Y \downarrow$. Fixed e : $Y \uparrow$.

32 (PS4.7). a. Upward-sloping LM. b. $e \uparrow, Y \uparrow$. c. $e \downarrow, Y \uparrow$.

33 (PS 4.8). $\frac{\partial Y}{\partial \pi^e} = \frac{1}{\Delta}[(I' + NFI')L_r] > 0$. $\frac{\partial r}{\partial \pi^e} = \frac{1}{\Delta}[(1 - C')L_r] < 0$. $\frac{\partial \epsilon}{\partial \pi^e} = \frac{NFI'}{NX'} \frac{1}{\Delta}[(1 - C')L_r] < 0$. $\Delta = -(1 - C')L_r - (I' + NFI')L_Y > 0$.

34 (PS5.1 ♡). a. $N^D = 25/(\frac{W}{P})^2$. b. $Y_n = 25$. c. $Y = 25P$.

35 (PS5.2). a. In the short run, $Y \downarrow, i \downarrow, P \downarrow, C \downarrow$; for $I(i - \pi^e)$, investment and private saving rise; for $I(i - \pi^e, Y)$, investment and private saving are ambiguous and the paradox may occur. b. In the long run, $Y = Y_n$. $i \downarrow, P \downarrow, C \downarrow$; investment and private saving rise. The paradox disappears.

36 (PS5.3). See the liquidity trap.

37 (PS5.4 ♡). It is possible.

38 (PS5.5 ♡). a. $(Y, P) = (1000, 6)$. b. $(Y, P) = (1600, 2)$.

39 (PS5.6). a. $\pi^e = 5, u = 5, L = 6.25$. b. $\pi^e = 0, u = 5, L = 5$. c. Recommend (b). d. $u = 2.5, L = 3.75$. e. Dynamic inconsistency.

40 (PS5.7). c. MP does not shift. IS shifts rightward. d. $\pi^* \downarrow$ shifts MP upward. In the short run, $Y \downarrow$ and $r \uparrow$. e. $\pi^e \uparrow$ shifts MP upward. In the short run, $Y \downarrow$ and $r \uparrow$. f. Yes, π^e will return to π^* over time. g. $\pi^e \uparrow$ shifts MP downward. $Y \uparrow, r \downarrow$ in the short run. h. No, π^e will not return to π^* . $a < 1$ is not sensible.

41 (PS 5.8). $\frac{\partial Y}{\partial G} = -\frac{1}{\Delta}L_i F_N > 0$. $\Delta = -(1 - C')F_N L_i + I'(F_{NN}M/W - L_Y F_N) > 0$.